

MP352 – Special Relativity

Problem Set 1

Due by the end of lecture on Friday, 6 March 2015

1. The FitzGerald-Lorentz explanation for the result of the Michelson-Morley experiment was that lengths in the direction of motion through the ether *contract*. More specifically, if L_0 is a length as measured on Earth, then the ether sees that length shrink to the value

$$L = \sqrt{1 - \frac{v^2}{c^2}} L_0$$

if it is parallel to the velocity $\vec{v}$ of the Earth through the ether.

This means that the Earth itself should be “squashed” into an oblate spheroid due to its motion. If we take $v = 30 \text{ km} \cdot \text{s}^{-1}$ (the Earth’s speed around the Sun), by how many centimetres would the Earth’s diameter contract in the direction of motion through the ether? (The speed of light is $c = 3 \times 10^8 \text{ m} \cdot \text{s}^{-1}$ and the Earth’s radius is 6371 km.)

2. If a frame $\mathcal{S}'$ moves at a constant velocity $\vec{v}$ with respect to an inertial frame $\mathcal{S}$, the general form for the coordinate transformation from $(t, \vec{r})$ to $(t', \vec{r}')$ is

$$\begin{aligned} t' &= \gamma(v) \left(t - \frac{\vec{v} \cdot \vec{r}}{c^2} \right), \\ \vec{r}' &= \vec{r} + \frac{\gamma(v) - 1}{v^2} (\vec{v} \cdot \vec{r}) \vec{v} - \gamma(v) \vec{v} t, \end{aligned}$$

where

$$v = |\vec{v}|, \quad \gamma(v) = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

- (a) Show that for any velocity $\vec{v}$,

$$-c^2 (t')^2 + \vec{r}' \cdot \vec{r}' = -c^2 t^2 + \vec{r} \cdot \vec{r}.$$

(In other words, do NOT assume that $\vec{v}$ points in any particular direction; show the above for *any* choice of the velocity vector.)

- (b) Show that if $\vec{v} = v \hat{e}_x$, you recover the usual transformations for a boost in the x -direction, i.e.

$$\begin{aligned} t' &= \gamma(v) \left(t - \frac{vx}{c^2} \right), & x' &= \gamma(v)(x - vt), \\ y' &= y, & z' &= z. \end{aligned}$$

3. A boost from one inertial frame to another can be characterised either by a speed v or a rapidity χ , with the two being related by

$$\cosh \chi = \gamma(v).$$

- (a) Show from this that the formula for getting the speed from the rapidity is

$$v = c \tanh \chi$$

and explain why this ensures that $|v| < c$.

- (b) By using $\cosh \chi = (e^\chi + e^{-\chi})/2$, confirm that the formula for getting the rapidity from the speed is

$$\chi = \frac{1}{2} \ln \left(\frac{1 + v/c}{1 - v/c} \right).$$

(In other words, plug the above expression into $\cosh \chi$ and show you get $\gamma(v)$.)

- (c) Use the above two expressions to calculate (i) the rapidity corresponding to a speed of $v = 10^8 \text{ m}\cdot\text{s}^{-1}$ and (ii) the speed corresponding to a rapidity of $\chi = 1.5$.